

Brain MRI Segmentation and Classification

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Abstract—Brain tumors are among the most life-threatening forms of cancer, and accurately diagnosing from a Magnetic Resonance Imaging (MRI) is critical for effective treatment planning. Manual analysis of MRI scans is time-consuming, subjective, difficult to scale and costly. This has driven growing interest in automated deep learning systems capable of analysing MRI scans with greater speed and consistency.

This project combines two deep learning models: U-Net and VGG16 for segmenting and localizing tumor regions and classifying the tumor into four categories: glioma, meningioma, pituitary tumor, and no tumor. The models were trained and evaluated on the BRISC 2025 Kaggle dataset, with preprocessing steps including resizing, normalization, and splitting into train/validation datasets. The U-Net segmentation model achieved a Dice coefficient of 0.824 and IoU of 0.750, while the VGG16 classifier achieved an overall accuracy of 94%.

Although the results are slightly lower than those reported by Arafat et al. [2], who achieved a Dice coefficient of 0.94 and classification accuracy near 99%, the proposed approach still demonstrates robust performance and highlights the effectiveness of combining segmentation and classification models for automated brain tumor analysis.

Index Terms—U-Net, VGG16, MRI, masks, labels, segmentation, classification.

I. INTRODUCTION

Brain tumors are serious health problems that need to be diagnosed correctly and quickly in order to get the best treatment. Magnetic Resonance Imaging (MRI) is a common way to find problems with the brain because it can make clear pictures of soft tissues. But analyzing MRI scans by hand takes a lot of time and can be different for each expert, which is why we need automated and reliable diagnostic systems. Recent improvements in deep learning, especially Convolutional Neural Networks (CNNs), have made medical image analysis tasks much better. U-Net-based architectures have become very popular for biomedical image segmentation because they can accurately find tumor areas and keep spatial information using encoder-decoder structures with skip connections [1].

Deep learning models like VGG16 have also shown that they can do a good job of classifying brain tumors by learning complex image features [8]. In this project, we create a combined pipeline that uses U-Net to segment tumors and a fine-tuned VGG16 model to classify them. The system finds tumor areas and sorts MRI scans into four groups: glioma, meningioma, pituitary tumor, and no tumor. We use metrics like Dice coefficient, IoU, accuracy, precision, recall, and F1-score to judge the models. This method shows how

well combining segmentation and classification works for automated brain tumor analysis.

Deep learning models have performed well, but there are still problems with accurately segmenting tumor boundaries and telling apart tumor types that look similar. Model performance can be affected by things like limited preprocessing, differences in datasets, and differences in training strategies. This project tackles these issues by looking at the results of the proposed pipeline and comparing them to other studies. It points out areas that need work and suggests ways to make brain tumor diagnosis more accurate and reliable.

II. PROBLEM STATEMENT

Manual brain tumor segmentation in MRI is a slow, labour-intensive task that depends on expert neuroradiologists and is difficult to scale across large patient volumes. In addition, tumor boundaries in MRI can be ambiguous due to low contrast, intensity inhomogeneity, and heterogeneous tumor appearance, which increases inter-observer variability and makes consistent lesion measurement challenging.

The problem addressed in this project is to develop and evaluate an automated, pixel-wise segmentation method that can accurately identify tumor regions in 2D brain MRI slices. Specifically, given an input MRI slice, the model must predict a binary mask that matches the ground-truth tumor annotation. The solution should generalize to unseen data and be assessed using segmentation-appropriate metrics (e.g., Dice and IoU) as well as qualitative visual comparisons between predicted and true masks. In addition, a classifying CNN will also be included in the pipeline to assist with labelling tumor into the correct category. The classification model will be assessed on metrics such as Precision, F1 and Recall scores, and the predicted labels will be compared to the ground truth ones.

III. LITERATURE REVIEW

A. U-Net: Convolutional Networks for Biomedical Image Segmentation

The paper introduces a type of Convolutional Neural Network called a UNet to segment different biological structures, such as cells and neuronal membranes in microscopy images. The architecture of a UNet can be broken down into two parts: contracting and expansive paths. The contracting path (left downward side) is responsible for compressing the image by using max-pooling. It follows a standard convolutional network design, progressively compressing the spatial dimensions

of the input through repeated 3×3 convolutions, ReLU activations, and 2×2 max-pooling operations, which together capture increasingly abstract contextual feature representations. The expansive path (right upward side) uses up-convolutions to progressively restore spatial resolution, ultimately producing a full segmentation map. In this process, the copy and crop connections are central to the design because they allow high-resolution spatial information from early layers to be combined with deeper, more abstract feature representations, enabling precise localization of segmented structures. The shape of this architecture resembles the letter 'U'; the contracting forming the downward path, the bottleneck at the base, and the expansive forming the upward path. This gives this convolutional neural network its name, UNet.

A significant advantage of this architecture is that U-Net can be trained to high accuracy with a limited amount of labelled data. This is achieved through data augmentation, where transformations such as shifting, rotation, and elastic deformations are applied to training images, allowing the network to learn robustness to the natural variations found in biological tissue.

The authors evaluated U-Net on two biomedical segmentation challenges: neuronal structure segmentation in electron microscopy stacks (ISBI 2012) and cell segmentation in light microscopy images (ISBI 2015). On the ISBI 2012 challenge, U-Net achieved a warping error of 0.0003529 and a rand error of 0.0382, surpassing previous best results. On the ISBI 2015 cell tracking challenge, U-Net won by a significant margin, achieving a mean IoU of 0.9203 on the PhC-U373 dataset and 0.7756 on the DIC-HeLa dataset. These results demonstrated that the architecture could produce highly accurate segmentations from very few annotated images, establishing U-Net as a foundational model for biomedical image analysis and motivating its widespread adoption in tasks such as brain tumor segmentation.

B. Brain Tumor MRI Image Segmentation and Classification based on Deep Learning Techniques

The paper uses a U-Net neural network to segment brain tumors from MRI images and a convolutional neural network architecture to classify the tumor-consistent images into glioma, meningioma, and pituitary tumors. The model is then assessed using classification accuracy and Dice coefficient measure. This approach relies on preprocessing, segmentation, and classification. For preprocessing, the authors use skull stripping to identify regions of interest. In MRI scans, skull stripping helps locate tumor tissues. Skin, skull, and other non-brain tissues are removed from the MRI. This produces a noise-free image for examination. U-Net is trained on brain MRI images with pre-labeled tumor areas. The model learns tumor region features from dataset photos and develops a binary mask to emphasize the input image's tumor region. Final tumor categorization uses a CNN. The model was trained using 3064 segmentation images and 3264 glioma, meningioma, and pituitary categorization images. U-Net divides brain images into areas and assigns cancer and normal tissue probabilities.

The classification CNN has 16 layers—13 convolutional and 3 fully connected. Each group of two or three convolutional layers has a max pooling layer. Segmentation model performance was Dice = 0.94, F1 = 0.76, Jaccard = 0.68, Recall = 0.74, Precision = 0.82, and classification model performance was per-class precision/recall/F1, with near 99% accuracy and good per-class F1 scores.

C. Automatic Brain Tumor Segmentation from MRI Scans using U-net Deep Learning

Brain tumor segmentation from MRI images is a challenge in the current medical landscape due to factors such as low contrast, uneven tumor forms, and scan intensity variation in MRI images. To address this issue the authors introduced an automated deep learning framework built on an advanced U-Net structure. The input images were preprocessed using Edge Enhancing Diffusion (EED) and region-of-interest (ROI) extraction. EED assists in improving the visibility of the tumor and ensures uniformity of pixel brightness, whereas ROI intersection reduces background noise and class imbalance. The proposed U-Net architecture, as seen in Figure 1, is an attention-based U-Net with Convolutional Block Attention Module (CBAM). It uses both spatial and channel attention methods to assist the network in filtering irrelevant features and concentrate on pertinent tumor locations. The model was tested on the BraTS 2020 dataset and it achieved a Dice coefficient of 90.8% and a sensitivity score of approximately 92%. The findings posted by the authors indicates that the combination of preprocessing techniques and attention-based U-Net significantly enhances the prediction of the model as compared to a base U-Net.

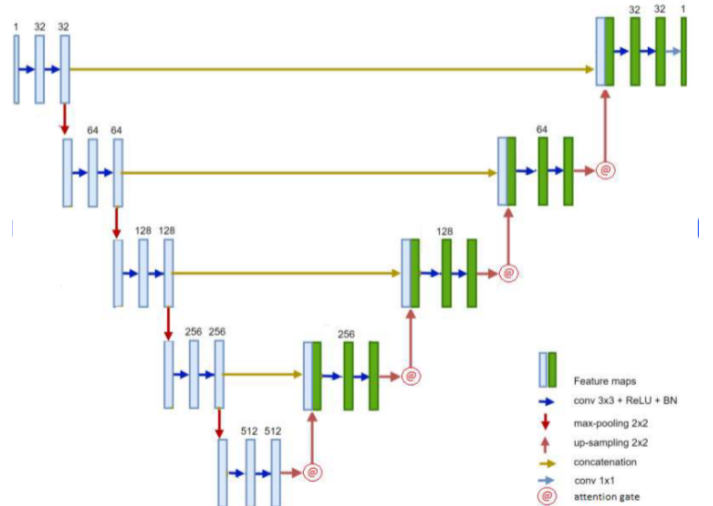


Fig. 1. Proposed U-Net Architecture [3].

D. Tumor Segmentation in Brain MRI: U-Nets versus Feature Pyramid Network

This paper compares three segmentation approaches for brain tumor MRI: standard U-Net, U-Net with a ResNeXt50 backbone, and a Feature Pyramid Network (FPN). The ResNeXt50 backbone enables deeper feature extraction within the U-Net framework, while the FPN builds multi-scale representations through a top-down pyramid with lateral connections and a sigmoid-activated output layer. Models are evaluated using Dice coefficient, loss, pixel accuracy, and IoU, with experiments run on Google Colab using an NVIDIA Tesla K80 GPU. With a split of 3005/393/531 images for train/val/test, the U-Net+ResNeXt50 model achieves the best performance with a Dice score of 0.932, loss of 0.142, pixel accuracy of 0.996 and IoU of 0.89, outperforming both the baseline U-Net and the FPN.

E. Brain Tumor Segmentation using Attention-based UNet

This study enhances U-Net by inserting the Convolutional Block Attention Module (CBAM) into skip connections so that encoder features are refined before being concatenated into the decoder. By applying channel and spatial attention, CBAM amplifies tumor-relevant features while suppressing background/healthy-tissue noise, improving the network's ability to localize heterogeneous tumors and preserve fine boundaries. Experiments on MRI datasets show the attention-augmented model consistently outperforms standard U-Net on common metrics such as Dice, IoU, and sensitivity, indicating more accurate and robust tumor segmentation. The authors also emphasize that attention improves interpretability by making the model's focus on relevant regions more explicit.

F. Visualizing UNet Decisions: An Explainable AI Perspective for Brain MRI Segmentation

This paper examines the decision-making of U-Net segmentation models for brain MRI by applying explainable AI methods (Grad-CAM, Guided Backpropagation, and Integrated Gradients) to visualize which image regions drive tumor predictions. The resulting activation maps generally align with expert-annotated tumor cores and boundaries, suggesting the model relies on clinically meaningful features rather than spurious cues. Grad-CAM is highlighted as providing the most interpretable high-level heatmaps, while Guided Backpropagation offers finer pixel-level detail. Overall, the work argues that integrating such XAI tools can improve trust, reveal biases or failure modes, and support safer clinical adoption of deep learning segmentation.

G. U-CCNet: Brain Tumor MRI Image Segmentation Model with Broader Global Context Semantic Information Abstraction

This paper extends U-Net with a context-enhancement module (U-CCNet) designed to strengthen edge details while also leveraging broader global semantic information. From backbone feature maps, the model projects features into query/key/value embeddings through 1×1 convolutions and

computes attention weights to aggregate context-aware representations, improving how each pixel is labelled using information from its surroundings. By fusing this CC-Net module with U-Net, the approach aims to reduce redundancy from noisy feature fusion and increase effective receptive field for more reliable boundary delineation. With an 80/10/10 train/val/test split, the model reports gains over standard U-Net, about 4.4% increase in Dice and 3.7% increase mean IoU, indicating more accurate tumor-region segmentation.

H. Image Processing based Brain Tumor Classification using VGG16 Compared with SVM to Improve Accuracy

The authors look to address the challenge of accurately classifying brain tumors from MRI images. The paper states that the traditional methods face issues in accuracy due to differences in image quality, scale, and pixel brightness. In addition, accuracy falls due to high levels of background noise and intra-class variability. Since early detection is very important for a patient's survival, the paper proposes the use of a deep learning model, the VGG16, and compares it with the SVM model, the traditional approach, to analyze if the accuracy improves or not. The VGG16 is a CNN model with 16 layers that utilizes 3 by 3 convolutional filters, 2 by 2 max pooling, and a fully connected layer with softmax for classification. The dataset used for training consisted of 30 sample images per group and the results achieved by the proposed VGG16 model outperformed the traditional SVM. In terms of accuracy and precision, the VGG16 achieved a score of 88.59% and 15.06, respectively. In contrast the SVM achieved an accuracy and precision of 84.58% and 13.70.

I. Multi-Class Brain Tumor Classification with MRI through a Fine-Tuned VGG16 Convolutional Neural Network

Manual diagnosis of brain tumor classification with MRI scans is an important issue as it's time-consuming and can result in false diagnoses with similarities between tumor and normal tissues. Because of this, studies are now applying deep learning techniques, like Convolutional Neural Networks (CNNs) for more accurate and mechanized classification and diagnosis. In the aforementioned paper, the authors suggested a transfer learning-based architecture with a fine-tuned VGG16 model, where pretrained layers are first frozen and then fine-tuned. They also suggested adding components such as global average pooling, batch normalization, dense layers, and dropout to enhance generalization and reduce overfitting. The model was practiced on a publicly available Kaggle dataset. This dataset had 7,023 MRI scans separated into four classes: glioma, meningioma, pituitary tumor, and no tumor, with preprocessing steps consisting of resizing, normalization, and data augmentation. The results in the paper showed a strong performance. The overall accuracy was 95% with high precision, recall, and F1-scores across all categories. The model performed best at distinguishing between tumors and non-tumors, though still some small confusion was present with structurally similar tumor types. In conclusion, this study showed the effectiveness of transfer learning-based CNN

models in bettering the accuracy and diagnosis of different brain tumor types.

J. Brain Tumor Segmentation and Classification Using Multi-Scale SAM and VGG16

Brain tumor diagnosis using MRI images can be problematic because of its time-consuming nature and because of the different tumor shapes, sizes, and unclear boundaries that can lead to false identification. The aforementioned paper proposed a two-staged deep learning pipeline as a way to resolve this issue. This pipeline combined segmentation and classification for improved performance. The suggested architecture first used a Multi-Scale and Detail-Enhanced Segment Anything Model (SAM) to accurately recognize and separate tumor regions, followed by a VGG16-based Convolutional Neural Network to arrange the segmented tumors into different categories. The model was trained and run through a dataset of 3,929 MRI scans, where preprocessing steps such as resizing to 224×224 and data augmentation techniques like rotation, flipping, and zooming were used to enhance generalization. The results demonstrated a strong performance. The segmentation model attained a 97% accuracy and an F1-score of approximately 96.7%, while the classification model attained a 98% accuracy in differentiating between different tumor types. The results of the paper showed that combining segmentation and classification into a single framework can significantly improve the accuracy and reliability of automated/mechanized brain tumor diagnosis systems.

IV. METHODOLOGY

A. Data Collection

The BRISC 2025 dataset from Kaggle contains both classification labels and segmentation masks for brain tumours. The dataset has been organized into four categories: glioma, meningioma, pituitary tumor, and no tumor. Alongside the classification labels, corresponding segmentation masks indicate the tumor regions in each scan. The dataset comes pre-split into training and test sets for both tasks.

The dataset provides a total of 6,000 labeled MRI scans for classification and 4,793 paired image-mask samples for segmentation, split across training and testing sets to enable robust model evaluation. Tables I and II provide a further breakdown of the dataset distribution across each split.

TABLE I
CLASSIFICATION DATASET DISTRIBUTION

Split	Class	Count
Train	Glioma	1,147
	Meningioma	1,329
	Pituitary	1,457
	No Tumor	1,067
	Total	5,000
Test	Glioma	254
	Meningioma	306
	Pituitary	300
	No Tumor	140
	Total	1,000

TABLE II
SEGMENTATION DATASET DISTRIBUTION

Split	Type	Count
Train	Images	3,933
	Masks	3,933
Test	Images	860
	Masks	860

B. Models Used

1) *U-Net*: To segment the tumour, the project utilizes the U-Net convolutional neural network. This architecture has two symmetric parts: an encoder and a decoder. The encoder (contracting path) is responsible for compressing the image using max-pooling, progressively reducing the spatial dimensions of the input through repeated 3×3 convolutions, ReLU activations, and 2×2 max-pooling operations, which together capture increasingly abstract contextual feature representations. The decoder (expansive path) uses up-convolutions to progressively restore spatial resolution, ultimately producing a full segmentation map. The copy and crop skip connections are central to the design, as they allow high-resolution spatial information from early layers to be combined with deeper, more abstract feature representations, enabling precise localization of segmented structures. In this project, the U-Net model is trained to segment tumour regions from brain MRI images by producing pixel-level predictions. For each input image, the model outputs a segmentation mask that highlights the location and shape of the tumour.

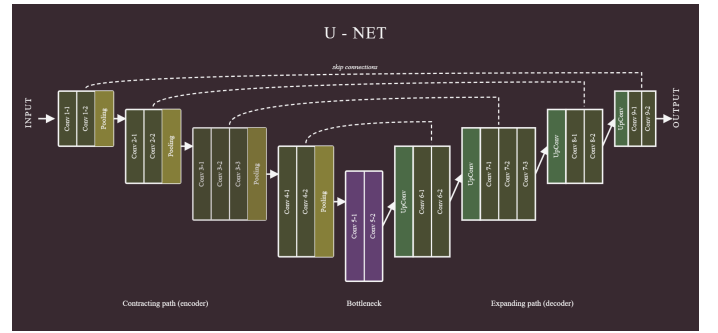


Fig. 2. UNet Architecture

2) *VGG16*: VGG16 (Visual Geometry Group, 16 layers) is a deep convolutional neural network developed by the Visual Geometry Group at the University of Oxford. It consists of 13 convolutional layers organized into five blocks, each followed by max-pooling, and three fully connected layers at the classifier head. For tumor classification, a pretrained VGG16 model was adopted using transfer learning, with weights loaded directly from PyTorch's `torchvision` repository pretrained on the ImageNet dataset. The convolutional base was frozen entirely during training, and only the final fully connected layer was replaced with a new linear layer mapping to four output classes: glioma, meningioma, no tumor, and pituitary.

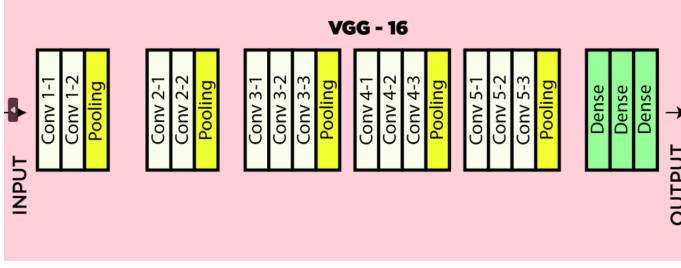


Fig. 3. VGG16 Architecture [11]

C. Data Preprocessing

Prior to model training, all MRI images were preprocessed using two separate pipelines to guarantee consistent input to each deep learning model. Both pipelines were implemented through a shared `preprocess_image()` function using the OpenCV Python library.

For the U-Net, the preprocessing pipeline first resizes each image to 224×224 pixels to ensure uniform input dimensions across the dataset. The images were then converted to grayscale, reducing each image from three channels (RGB) to one, which is appropriate for MRI scans where pixel values represent tissue intensity rather than colour information. Pixel values were then normalized by dividing by 255.0, scaling the range from $[0, 255]$ to $[0.0, 1.0]$. Normalization ensures stable gradient updates during backpropagation. Finally, a channel dimension of size 1 was explicitly added to produce a final tensor shape of $(224, 224, 1)$, as required by the Keras `ImageDataGenerator` and the U-Net input layer.

For the VGG16, the preprocessing pipeline similarly resizes each image to 224×224 pixels to match the expected input dimensions of the pretrained model. The colour channels were rearranged from BGR to RGB using OpenCV, followed by a type conversion to float. Since VGG16 was pretrained using mean subtraction normalization, where the ImageNet channel means are subtracted from each pixel across the Red, Green, and Blue channels, Keras provides the `preprocess_input()` function to apply this automatically, ensuring the input distribution of the MRI images is consistent with what the pretrained VGG16 weights expect.

D. Experimental Metrics

The performance of both models was measured using metrics appropriate to each task. For the U-Net, segmentation accuracy was assessed by measuring the overlap between the predicted mask and the ground truth mask using two commonly used metrics: the Dice coefficient and Intersection over Union (IoU), also known as the Jaccard index, defined as follows:

$$\text{IoU} = \frac{|A \cap B|}{|A \cup B|} \quad (1)$$

$$\text{Dice} = \frac{2|A \cap B|}{|A| + |B|} \quad (2)$$

A denotes the predicted segmentation mask output by the U-Net model and B denotes the ground truth mask representing the manually annotated tumor region.

The Dice coefficient measures the overlap between two binary masks by computing twice the shared area divided by the total combined area, producing a score between 0 and 1 where 1 indicates a perfect match. IoU also quantifies overlap, but divides the intersection by the union of the two masks, producing a more conservative score.

For the VGG16 classification model, performance was evaluated using overall accuracy, defined as the proportion of correctly predicted labels across the four tumor categories.

V. RESULTS AND ANALYSIS

The U-Net VGG16 pipeline for segmentation and classification was trained and tested on the BRISC 2025 dataset available on Kaggle, which contains both segmentation masks and classification labels. The U-Net portion of the pipeline was trained on 3539 images and validated on 394 images, whereas VGG16 was trained on approximately 5700 images and tested on 1300 images across the four tumor classes: glioma, meningioma, no_tumor, and pituitary. The U-Net segmentation model achieved strong performance on the MRI segmentation task for the brain, as after training it achieved a loss of 0.0186 with a binary accuracy of 0.9982. The U-Net model also achieved an IoU of 0.7503 and a Dice coefficient of 0.8237. These results indicate that the model was successfully able to learn the tumor region boundaries and was able to produce prediction masks that closely follow the ground truth mask, as seen in Figure 4.

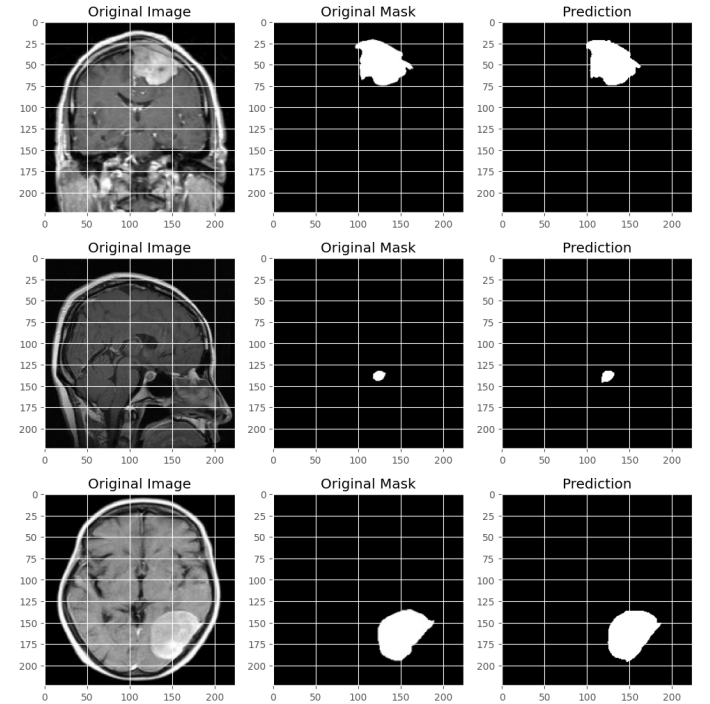


Fig. 4. U-Net Prediction Mask Compared to Ground Truth.

Inspecting Figure 4 closely, it can be seen that the model’s prediction mask is able to capture the main shape of the tumor in most cases, and it follows the general structure of the ground truth, being able to differentiate the brain tissues from the tumor. This success is because of the U-Net’s encoder-decoder architecture with skip connections between the two, helping the model to preserve contextual and spatial information. However, since the recorded IoU and Dice coefficient are being recorded at 0.7503 and 0.8237 respectively, it does suggest that there is room for improvement near the edges of the tumor and in areas with less clarity where it’s harder to obtain accurate results at the pixel level.

For the classification part of the pipeline, a pretrained VGG16 model was used and fine-tuned further to classify the MRI images into four classes: glioma, meningioma, no_tumor, and pituitary. Over the course of 10 epochs, the model was trained. From the first epoch’s 0.7005 accuracy to the last epoch’s 0.9677 accuracy, the training loss went down from 1.6444 to 0.1000. This steady growth shows that the model learned how to tell the difference between the MRI scans correctly. On the held-out evaluation split, the VGG16 model got an overall accuracy of 0.93, which means it did a satisfactory job of classifying all four types of tumors. The results of the VGG16 on the test set can be seen in Figure 5.

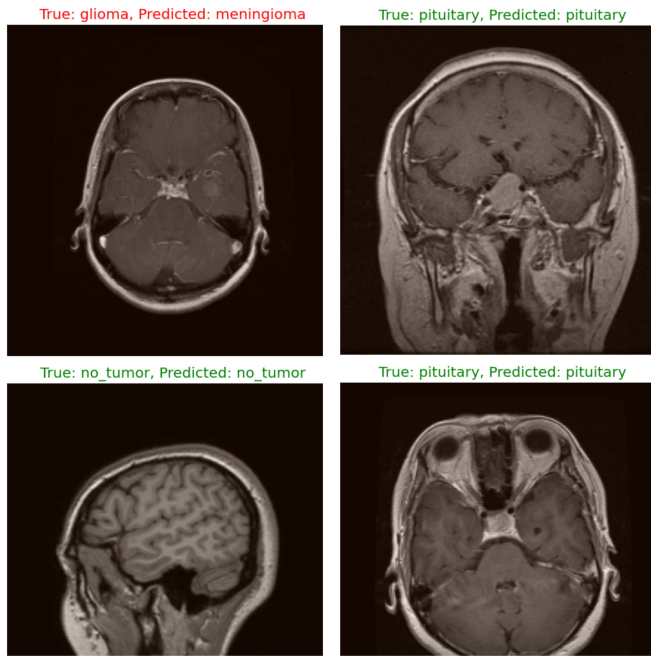


Fig. 5. VGG16 Predicted Label Compared to Ground Truth.

With F1-scores of 0.98 and 0.96, the model did best on the no_tumor and pituitary classes, according to a more in-depth class-wise study from the classification report. It did very well in the glioma class as well, with an F1-score of 0.90, an accuracy of 0.89, and a recall of 0.92. The hardest class was meningioma, with an F1-score of 0.87 and a memory of 0.83, which means that some images of meningiomas got

mixed up with images of other types of tumors. Still, both the macro-average and weighted-average F1-scores were 0.93, which showed that the classifier was fair and didn’t rely too much on just one class.

Comparing the results from the paper “Brain Tumor MRI Image Segmentation and Classification based on Deep Learning Techniques”, the authors reported a Dice coefficient of 0.94 and a Jaccard index of 0.68. In contrast, the U-Net model in this project achieved a Dice coefficient of 0.82 and an IoU, same as Jaccard index, of 0.75. There was a difference in performance because of things like different preprocessing steps, such as skull stripping and ROI extraction used in the study, different datasets, and maybe fewer training epochs or easier model tuning in the implementation. In terms of classification, the authors reported that both training and validation were almost 100% accurate and that F1-scores for each class were very high, ranging from 0.98 to 0.99. The VGG16 model in the project, on the other hand, got an average accuracy of 93%, though it did a little worse on the meningioma class. This gap shows that the model did a good job of generalizing, but it did not do as well as the reference work. One possible reason for this difference is that the paper used a custom CNN design that was trained on split-up tumor regions, while this project used a VGG16 model that had already been trained but may not have been fully fine-tuned. Differences in dataset size, class distribution, and training strategy (e.g., epochs, augmentation, or hyperparameter tuning) may have affected the final accuracy.

Overall, the project and the reference work show that using both U-Net and CNN-based classification together is a good way to look at brain tumors. The reference study, on the other hand, got better results with segmentation and classification. This suggests that better preprocessing, longer training, or model optimization could make this project’s implementation work better.

VI. CONCLUSION

In conclusion, the aim of the project was to build a U-Net VGG16 pipeline that would be capable of tumor segmentation and classification and compare the results with the paper “Brain Tumor MRI Image Segmentation and Classification based on Deep Learning Techniques.” The project utilized the BRISC 2025 dataset, which included train and test images for both segmentation and classification tasks. The images went through a preprocessing stage where the MRI images were resized to 224 pixels in width and height, normalized, and structured into consistent input formats, which helps in stabilized training and improved convergence. Normalization fixed the values of pixel intensities so they were scaled correctly. This made changes in picture brightness and contrast less noticeable. Putting the data into the right training, validation, and testing groups also stopped data from getting lost and made sure that the model’s performance could be accurately evaluated. In this project, the preprocessing wasn’t as complex as the reference paper with methods like head stripping or

region-of-interest extraction, but it was still enough for the models to learn useful features from the data.

The U-Net model was successful in detecting the tumor regions, achieving a Dice coefficient and IoU score of 0.8237 and 0.7503 respectively, which indicates a good overlap between the model prediction and ground-truth masks. The VGG16 classification model achieved an accuracy of 93% across the four classes, which were glioma, meningioma, pituitary, and no tumor indicating the model's effectiveness in distinguishing between different types of tumors.

The project's results were slightly lower than the reference study's, especially in terms of how well it did at segmentation and classification. This difference can be explained by different ways of preprocessing, setting the model, and training it. The higher Dice score (0.94) and classification accuracy (99%) of the reference paper were due to its use of more advanced preprocessing and probably better optimized training configurations. In spite of this, the project's results still show that the suggested pipeline works and can make accurate predictions.

Overall, the project shows that using a single deep learning system for both segmentation and classification is a powerful way to automate brain tumor analysis. Further refinements could include adding more advanced preprocessing methods like skull stripping, data augmentation, fine-tuning deeper layers of VGG16, and testing out more advanced U-Net variants. These improvements could make segmentation and classification even more accurate, bringing the findings closer to the state-of-the-art models.

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